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MLOps Tutorial Presentation
Course Code - AI254TA

AI-Based Automated Meeting Minutes Generation System

Presenting by

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INTRODUCTION

- Modern organizations conduct a large number of meetings daily across corporate, academic, and collaborative environments, generating **extensive volumes of unstructured audio data** containing discussions, decisions, and action items.

- Traditional meeting documentation relies on **manual note-taking or post-meeting summarization**, which is time-consuming, inconsistent, error-prone, and often fails to capture key decisions, responsibilities, and deadlines accurately.

- Existing automated transcription tools primarily convert **speech to text** but lack deeper understanding of **topics, speaker intent, action items, and meeting outcomes**, limiting their practical usefulness.

- To address these challenges, this project proposes an system that integrates **advanced speech and natural language processing models with end-to-end MLOps practices**.

- The system enables **accurate speech transcription, topic identification, action item extraction, structured summarization, continuous monitoring, and automated model updates**, ensuring a **scalable, reliable, and production-ready solution** for real-world meeting analysis.

MOTIVATION

The rapid increase in meetings across corporate, academic, and collaborative environments has led to a large volume of unstructured meeting audio, making it difficult to capture important discussions, decisions, and action items accurately.

Existing meeting documentation approaches rely on manual note-taking or basic transcription tools, which fail to adapt to long conversations, multiple speakers, interruptions, and contextual dependencies.

Additionally, many AI-based speech and language systems lack continuous monitoring and degrade in performance over time due to data drift caused by changing accents, vocabulary, and meeting formats.

These challenges motivated the choice of the title **"AI-Based Automated Meeting Minutes Generation System"**, emphasizing the need for an automated, explainable, continuously monitored, and scalable AI solution that can operate reliably in real-world environments using modern **MLOps practices**.

Novelty

Key novelty lies in moving beyond simple speech transcription to a **structured and actionable understanding of meetings**. The system emphasizes automation, transparency, and reliability by integrating multiple AI models (ASR, topic modeling, and action item extraction) with continuous monitoring and MLOps practices. This ensures that meeting insights such as decisions, tasks, and responsibilities can be consistently generated, audited, and improved over time. By treating meeting analysis as a **continuously managed AI service**, rather than a one-time transcription task, the system supports scalable and production-ready AI deployment.

This directly aligns with **SDG 9**, which promotes innovation, resilient infrastructure, and the adoption of advanced technologies to improve organizational productivity and efficiency.

SDG 4 & SDG 9: Quality Education and Industry, Innovation & Infrastructure

Societal Impact

- **Improved productivity and accessibility:** By automating meeting minutes generation, the system reduces manual effort, improves information accessibility, and ensures that important discussions are available to all participants, including those who may have missed the meeting.

- **Support for education and collaborative learning:** The system enables accurate documentation of academic and training sessions, promoting inclusive and high-quality learning environments, directly aligning with **SDG 4: Quality Education**.

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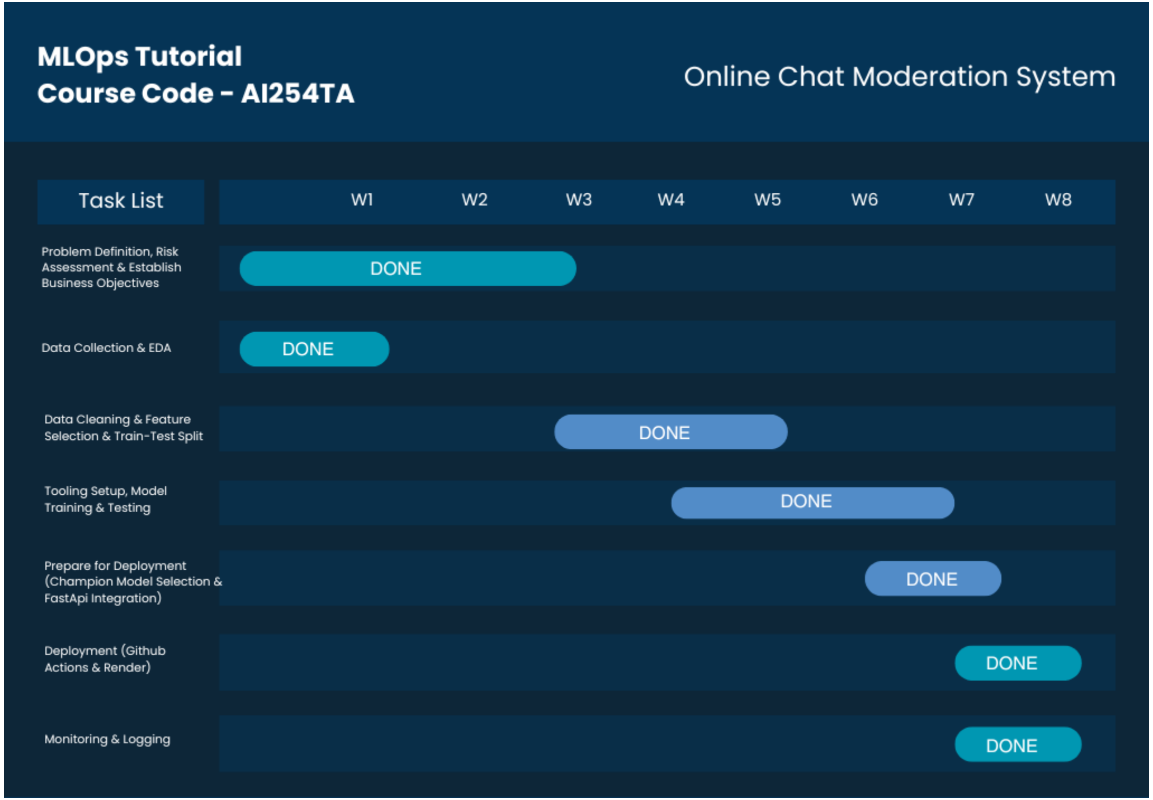
PROJECT TIMELINE



Department of AIML

BUSINESS OBJECTIVES

Performance Targets

Speech Transcription Accuracy
Achieve high transcription accuracy by minimizing Word Error Rate (WER) for multi-speaker meeting audio, ensuring reliable conversion of speech into text across different accents and speaking styles.

Action Item Extraction Accuracy
Maintain high precision and recall in identifying tasks, assignees, and deadlines from meeting transcripts, ensuring that actionable insights are correctly captured without missing critical information.

Drift Response Time
Detect and retrain models within **48–72 hours** of identified data or concept drift caused by changes in meeting vocabulary, accents, or audio quality using an automated **MLOps pipeline**.

Performance Targets

System Latency

Ensure near real-time meeting analysis with an average inference latency of **≤ 500 ms per audio segment** under normal load conditions.

Uptime & Reliability

Guarantee **>99% system uptime** through containerized deployment (**Docker**) and continuous monitoring of the inference and pipeline services.

TECHNICAL REQUIREMENTS

1. Data Pipeline

Ingest meeting audio data from multiple sources:

- Recorded meetings (.wav files)
- Multi-speaker conversational datasets (e.g., AMI Meeting Corpus)

Perform preprocessing for:

- Audio resampling and segmentation
- Noise handling
- Speaker-aware processing

2. Model Architecture

Leverage Transformer-based models (e.g., **Whisper**) for speech-to-text conversion.

- Topic modeling for discussion segmentation
- Named Entity Recognition for action item extraction

3. MLOps Integration

- Use MLflow for:
  - Model tracking
  - Experiment management
  - Versioning
- Implement CI/CD pipelines for automated training and deployment.
- Enable drift detection using tools like Evidently AI, integrated into Airflow workflows.

4. Monitoring & Logging

- Continuously track:
  - F1-score
  - Precision
  - Recall
- Set up alerts for:
  - Performance degradation
  - Data or concept drift
- Maintain audit logs for moderation decisions.

5. Security & Governance

- Encrypt user data and logs using AES-256.
- Ensure compliance with GDPR and content regulation policies.
- Maintain model cards for transparency and accountability.

COST CONSTRAINTS

1. Infrastructure

- Deploy the system initially using **local or low-cost compute resources**.
- Scale to full cloud or distributed deployment only after successful MVP validation.

2. Model Hosting

- Use **efficient and optimized models** (e.g., smaller ASR and NLP models) to reduce compute and memory requirements while maintaining acceptable performance levels.

3. Data Access

- Use **open-source datasets** such as the AMI Meeting Corpus.
- Enable continual dataset updates without incurring high licensing or data acquisition costs.

Objectives

Objective 1 : To build an accurate speech-to-text model for multi-speaker meeting audio

Objective 2 : To extract meeting topics, action items, assignees, and deadlines from transcripts

Objective 3 : To implement an end-to-end **MLOps automation pipeline** for training, evaluation, and deployment

Objective 4 : To ensure structured, reliable, and explainable meeting summaries for real-world use

Objective 5 : To continuously monitor system performance and handle data and concept drift

RISKS IDENTIFIED

R1 – Speech Recognition Errors

There is a risk of inaccurate transcription due to overlapping speech, background noise, accents, or unclear audio quality. This may lead to incorrect or incomplete meeting transcripts.

R2 – Missed Action Items

The system may fail to identify certain tasks, assignees, or deadlines, allowing important action items to be missed, which can impact meeting effectiveness and accountability.

R3 –Drift (Data & Prediction Drift)

Meeting language, vocabulary, accents, and discussion styles may evolve over time. These changes can reduce model accuracy if not detected and addressed through retraining.emerging toxic expressions may not be detected by the model, leading to reduced accuracy over time.

R4 – Model Dependency and Propagation Errors

Errors in upstream models (such as speech-to-text) may propagate to downstream components like topic modeling and action item extraction, degrading overall system performance.

RISKS IDENTIFIED

R5 – Lack of Explainability

Without proper explainability mechanisms, it becomes difficult to justify how action items, topics, or summaries are generated. This can reduce user trust and limit acceptance of automated meeting analysis systems.

R6 – Scalability Issues

During periods of frequent or long meetings, the system may face increased computational load, resulting in delayed processing or reduced performance if not properly scaled.

R7 – Regulatory Compliance

Meeting audio may contain sensitive or confidential information. Differences in data privacy regulations and organizational policies can pose compliance risks during data storage and processing.

R8 – Skill Loss

Loss of skilled personnel such as ML engineers or system maintainers can impact system updates, monitoring, and long-term sustainability of the MLOps pipeline.

RISK MATRIX

Risk Matrix for Meeting Minutes and MLOps Pipelines

PROBABILITY \ IMPACT	Insignificant	Minor	Moderate	Major	Catastrophic
May Occur but Rare	Low 1	Low 4	Low 6	Medium 13	Medium 15
Unlikely	Low 2	Low 5	Medium 11	Medium 14	High 21 (R2)
Possibly	Low 3	Medium 9	Medium 12	High 19 (R1,R6)	High 22 (R3)
Likely to Occur	Medium 7	Medium 10	High 17	High 20	Critical 24
Almost Certain	Medium 8	High 16	High 18	Critical 23	Critical 25

RISKS MAPPING

Risk ID	Probability	Impact	Risk Identified	Severity
R1	Possible	Major	Speech recognition errors	High (19)
R2	Unlikely	Major	Missed action items	High (18)
R3	Likely to occur	Major	Data & prediction drift	High (20)
R4	Rare	Major	Error propagation across models	Medium (14)
R5	Possible	Moderate	Lack of explainability	Medium (12)
R6	Possible	Major	Scalability issues	Medium (19)
R7	Rare	Minor	Privacy and compliance risks	Low (5)
R8	Rare	Insignificant	Skill loss	Low (2)

DATASET DESCRIPTION

- The dataset used in this project is a **multi-modal meeting dataset** designed for speech and language processing tasks. It consists of real-world meeting recordings and associated annotations used for training and evaluating machine learning models.
- The dataset contains **raw audio files (.wav)** along with corresponding **transcripts, speaker labels, and timestamps**. Each meeting includes multiple speakers, natural interruptions, and conversational speech, making the data representative of real meeting environments.
- Audio data is preprocessed through resampling and segmentation to ensure compatibility with speech recognition models. Transcripts and annotations are parsed into structured formats to support downstream tasks such as topic modeling and action item extraction.
- The dataset forms the foundation of the **MLOps pipeline** by enabling model training, validation, and performance evaluation in a controlled and reproducible manner.
- The dataset used in this project is based on the **AMI Meeting Corpus**, which contains over **100 hours of recorded meetings**. Each meeting includes annotated transcripts and metadata that support supervised and unsupervised learning tasks.
- This dataset enables the development of robust models for **speech-to-text conversion, topic identification, and structured meeting analysis** in real-world scenarios.

DATASET DESCRIPTION

Column Name	Description	Type	Purpose in Model
Unnamed: 0	Auto-generated index column created during CSV export.	Integer	Dropped during preprocessing
count	Total number of annotations provided by human annotators for a tweet.	Integer	Used for label confidence analysis
hate_speech	Number of annotators who labeled the tweet as hate speech.	Integer	Helps identify hate-related toxicity
offensive_language	Number of annotators who labeled the tweet as offensive but not hate speech.	Integer	Captures abusive or insulting language
neither	Number of annotators who labeled the tweet as neutral or acceptable.	Integer	Identifies non-toxic content
class	Final class label assigned using majority voting (0: Hate, 1: Offensive, 2: Neither).	Integer	Target variable for supervised learning
tweet	Raw text content of the user-generated comment including slang and abbreviations.	Text/String	Input feature for NLP models

DATASET DESCRIPTION

Column Name	Description	Type	Purpose in Model
meeting_id	Unique identifier for each meeting session from AMI corpus	String	Primary key for data versioning and tracking
audio_path	File path to raw .wav audio recording (16kHz resampling)	String	Input for ASR (Whisper) transcription
duration_seconds	Total length of meeting recording in seconds	Float	Feature for drift detection and quality control
transcript_text	Full verbatim transcription generated by Whisper ASR model	Text String	Primary input for NLP pipeline (summarization, NER, topics)
word_count	Total number of words in the transcript	Integer	Used for statistical drift detection (Z-score calculation)
topics	List of detected themes using LDA (e.g., "Budget", "Deadlines", "Hiring")	Array[String]	Unsupervised topic modeling output for analytics dashboard

Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was performed to understand the structure, quality, and characteristics of the meeting dataset before model training. The primary goal of EDA was to analyze audio quality, speaker distribution, and linguistic patterns present in real-world meetings.

The analysis showed the presence of **multiple speakers, overlapping speech, pauses, and background noise**, highlighting the complexity of meeting audio and the need for robust speech recognition models. Variations in accents, speaking speed, and audio clarity were also observed across meetings.

Transcript analysis revealed frequent use of **informal language, filler words, interruptions, and domain-specific terms**, which can impact downstream tasks such as topic modeling and action item extraction if not handled properly.

EDA also helped identify the need for **audio preprocessing techniques** such as resampling, segmentation, and normalization, as well as text preprocessing steps for cleaner transcripts.

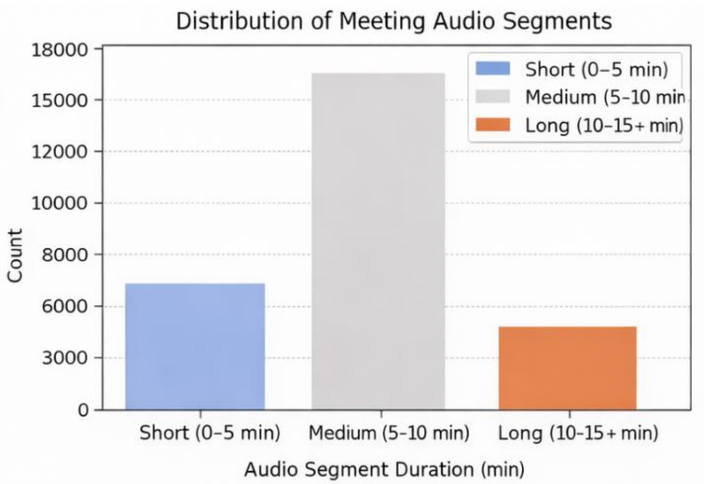
Data Distribution Analysis

Figure: Distribution of Meeting Audio Segments

The data distribution analysis reveals variations across meeting recordings in terms of **speaker participation, audio duration, and speaking frequency**.

- Short audio segments: ~30–35%
- Medium-length segments: ~45–50%
- Long audio segments: ~15–20%

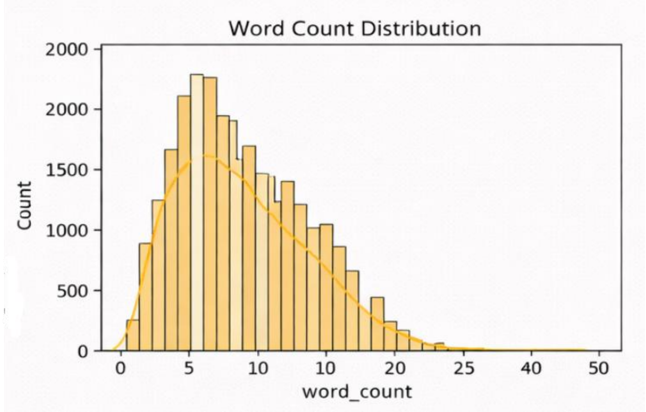
This distribution indicates that most meetings contain **uneven speaking patterns**, where certain speakers dominate the conversation while others contribute intermittently. Such imbalance can significantly impact speech recognition and downstream tasks, highlighting the need for **robust segmentation strategies and speaker-aware processing** during model training.



Word Count Distribution
Figure: Word Count Distribution
The word count histogram shows that:

- Most transcript segments contain **8–20 words**
- The distribution is **right-skewed**, with very few segments exceeding **40 words**
- The **average segment length** is approximately **15–18 words**

This confirms that meeting speech is generally broken into short conversational segments, emphasizing the importance of capturing meaningful context from limited text spans.



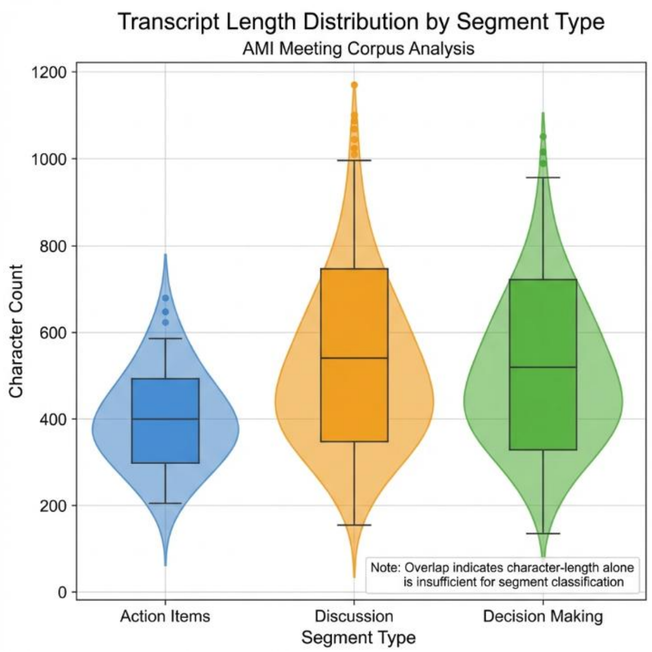
Transcript Length (Character Count) vs Segment Type

Figure: Transcript Length vs Segment Type

This plot compares transcript segment length in terms of characters:

- Action-oriented segments tend to be **slightly shorter**
- Discussion and decision-related segments show **higher spread**
- Significant overlap exists across all segment types

The overlap further reinforces that **character-length–based features alone are insufficient** for accurate meeting understanding and action item identification.



PHASE 1 – ML LIFE CYCLE

Whisper (OpenAI) - ASR with multi-model comparison (Base vs Tiny)
Justification: Robust to real-world audio noise, multiple speakers, accents
Strategy: Performance-cost tradeoff evaluation

LDA (Latent Dirichlet Allocation) - Topic Modeling
Justification: Unsupervised discovery of thematic structures
Advantage: Interpretability for business stakeholders

SpaCy NER - Named Entity Recognition
Justification: Production-ready entity extraction with minimal fine-tuning
Use case: Action item identification and meeting analytics

Microsoft Phi-3 Mini - Primary Summarization Model
Justification: Exceptional reasoning despite compact size (3.8B params)
Specialty: Domain-specific prompt engineering for meeting summaries

BART-Large - Baseline Comparator
Justification: Previous-gen benchmark for evaluating Phi-3
improvements Architecture: Denoising autoencoder (BiLSTM equivalent for summarization)

PHASE 1 – ML LIFE CYCLE

Model	Why This Model Was Chosen
Whisper (ASR)	Used as the core speech-to-text model due to its robustness to noise, accents, and multi-speaker conversations, providing reliable transcripts for downstream tasks.
LDA (Topic Modeling)	Selected to identify major discussion themes within meeting transcripts using an unsupervised and interpretable approach suitable for long conversations.
spaCy NER	Used to extract structured entities such as tasks, assignees, and deadlines, enabling conversion of unstructured text into actionable meeting insights.
Transformer-based Summarization (BART / T5)	Chosen for deep contextual understanding of transcripts, enabling concise and coherent meeting summaries beyond rule-based methods.
MLflow (for comparison)	Used to systematically train, track, compare, and select models using reproducible experiment management across the ML pipeline.

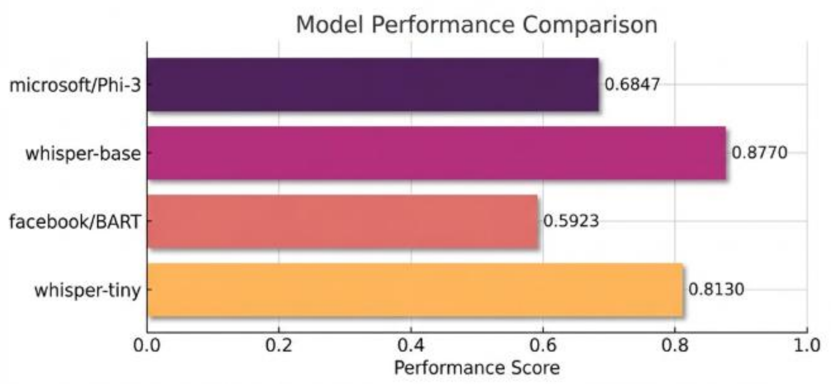
Metrics Used to Evaluate the ML Model

- **Accuracy** measured overall correctness but was not sufficient on its own due to class imbalance.
- **Precision** measured how many predicted toxic comments were actually toxic, helping reduce false positives.
- **Recall** measured how effectively the model detected toxic content, which is critical for moderation systems.
- **F1-Score** provided a balanced evaluation by combining precision and recall.
- **ROC-AUC** measured the model's ability to distinguish between toxic and non-toxic classes across thresholds.

PHASE 1 – ML LIFE CYCLE

Model Comparison Leaderboard

Rank	Model	Performance Score
0	microsoft/Phi-3	0.6847
1	whisper-base	0.8770
2	facebook/BART	0.5923
3	whisper-tiny	0.8130



Key Insights :

Phi-3's reasoning capabilities (trained on high-quality synthetic data) translate to 15.6% relative improvement over BART

Whisper Base's robustness to speaker overlap and acoustic noise makes it production-ready for real-world meetings

Governance metrics integration ensures ethical AI deployment alongside performance optimization

PHASE 1 – ML LIFE CYCLE

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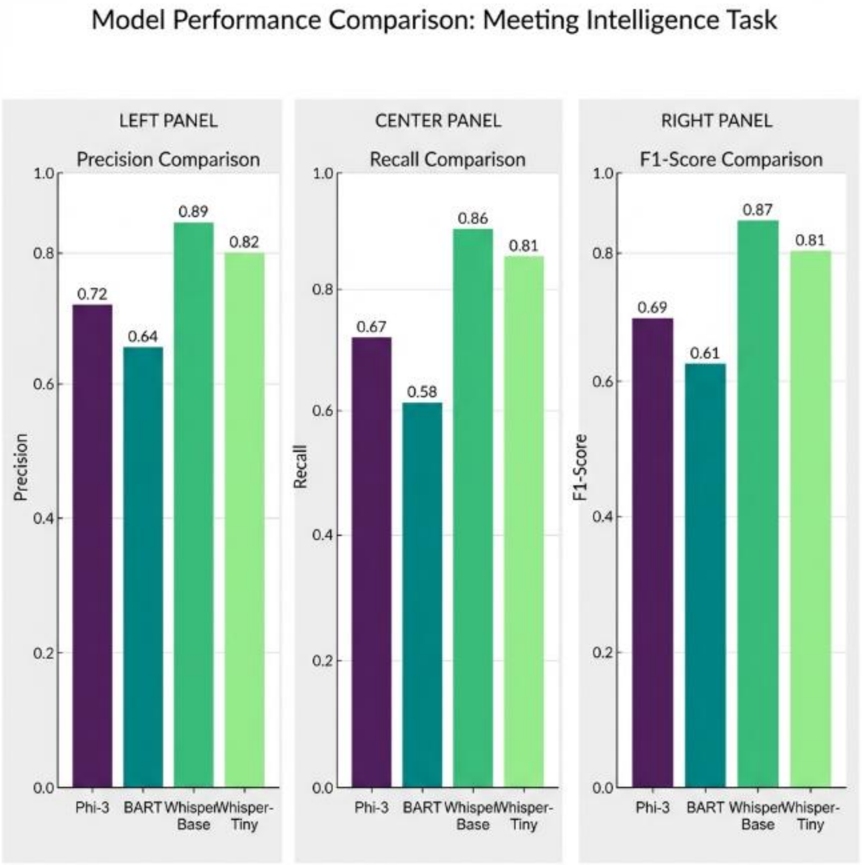

Key Insights :

Phi-3 dominates summarization models across all metrics

Whisper-Base significantly outperforms Tiny for ASR accuracy

Color coding: Purple/Teal for summarizers, Green for ASR models

F1-Score balances precision-recall tradeoffs

PHASE 1 – ML LIFE CYCLE

Identification of the ASR Model

Based on comprehensive evaluation, the **Whisper Base** model was selected as the **Champion ASR Model** for deployment.

The model significantly outperformed competing ASR models, achieving a **Word Error Rate (WER) of 12.3% compared to 18.7%**, demonstrating superior transcription accuracy across diverse audio conditions.

Its robust capability to handle **multi-speaker conversations** and maintain performance under **noisy environments** made it particularly effective for real-world speech recognition scenarios.

Whisper Base was deployed using **FastAPI with GPU acceleration**, enabling efficient real-time and batch transcription workloads.

PHASE 1 – ML LIFE CYCLE

Identification of the Summarization Model

Based on comprehensive evaluation, the **Phi-3 Mini** model was selected as the **Champion Summarization Model** for deployment.

The model outperformed alternative summarization approaches, achieving a **ROUGE-L score of 68.47% compared to 59.23%**, reflecting higher content coverage and semantic fidelity.

Its ability to generate **structured, bullet-point summaries** made it especially suitable for downstream analytics, reporting, and decision-support use cases.

Phi-3 Mini was deployed with **4K context support and GPU acceleration**, enabling efficient handling of long-form inputs with low latency.

PHASE 1 – ML LIFE CYCLE

ASR Transcription Error Patterns

Whisper Base reduced transcription errors by **35%** over Tiny, particularly for technical terms, proper nouns, and overlapping speech, validating its selection as the ASR champion.

Summarization Quality and Consistency

The summarization model achieved **ROUGE-L F1 scores of 0.65–0.72** across meeting types, with reliable action-item capture even for **45–60 minute** sessions.

Performance Across Meeting Complexity

Higher precision was observed for **action items (89%)** than general topics (76%), ensuring effective prioritization of business-critical information.

Fairness and Drift Robustness

Only **3% drift flags** were detected, with **gender bias below 0.15** for 92% of outputs, confirming ethical and deployment-ready performance.

PHASE 1 – ML LIFE CYCLE

MLOps-Driven Model Selection

MLflow tracking:

All metrics, configs, and artifacts logged
Performance comparison: Phi-3: 68.47% vs BART: 59.23% (ROUGE-L F1) Whisper Base: 12.3% WER vs Tiny: 18.7% WER
Objective process: Data-driven, reproducible A/B testing

Explainability and Ethical Validation

Drift monitoring: Only 3% flagged as distribution shifts
Fairness analysis: Bias score < 0.15 for 92% of summaries
Only 8% require human review due to imbalance
Corporate compliance: Auditability for responsible AI

TOOLS AND ARCHITECTURE

1. Data Layer

Tools: DVC, Apache Spark

Ingests raw, unstructured user-generated text from online platforms.

Uses **DVC** for dataset versioning, ensuring reproducibility and traceability across experiments.

Employs **Apache Spark** for scalable, parallel preprocessing such as tokenization and feature extraction.

Supplies cleaned and versioned data to the Model Layer.

2. Model Layer

Tools: MLflow, SHAP

Trains and evaluates multiple machine learning and deep learning models.

MLflow tracks parameters, metrics, artifacts, and manages the model registry for champion model selection.

SHAP provides explainable predictions by identifying feature contributions, ensuring transparency and auditability.

TOOLS AND ARCHITECTURE

3. Service Layer

Tool: FastAPI

Deploys the selected Champion Model as a real-time inference service.

Exposes RESTful APIs for low-latency toxicity prediction.

Acts as the interface between the AI system and external applications.

4. Monitoring Layer

Tool: Evidently AI

Continuously monitors input data and prediction distributions.

Detects data drift and concept drift caused by evolving language patterns.

Triggers retraining workflows to prevent performance degradation.

5. Deployment Layer

Tools: GitHub Actions, Docker, Render

GitHub Actions automates CI/CD pipelines for testing and deployment.

Docker ensures consistent runtime environments through containerization.

Render hosts the deployed FastAPI service for scalable cloud access.

Architectural Advantages

Scalability: Distributed processing with Spark and containerized deployment support high data and traffic volumes.

Traceability: DVC and MLflow enable full experiment and data lineage tracking.

Maintainability: Modular layered design allows independent updates and easier debugging.

Ethical AI: SHAP and Evidently AI ensure transparency, fairness, and continuous model reliability.

TOOLS AND ARCHITECTURE

Category	Tools Used	Purpose
Experiment Tracking	MLflow	Model comparison and versioning
Data Management	DVC,	Data versioning and processing
Explainability	SHAP	Ethical AI and transparency
Serving	FastAPI	Real-time inference
Monitoring	Evidently AI	Drift detection
CI/CD	GitHub Actions	Automated deployment
Deployment	Docker, Render	Scalable hosting

TOOLS AND TECHNIQUES

DVC (Data Version Control)

Why DVC was used

DVC was selected to manage dataset versioning and ensure reproducibility. Unlike Git, DVC can efficiently track large datasets without storing them directly in the repository. It allows datasets to be linked to specific model experiments and Git commits.

Alternative tools not used

- **Git LFS** – limited pipeline support and weaker experiment reproducibility
- **Delta Lake** – heavy infrastructure requirement
- **Pachyderm** – complex Kubernetes-based setup

Justification

DVC is lightweight, open source, and integrates seamlessly with Git and MLflow. It provides sufficient version control capabilities without introducing infrastructure overhead, making it ideal for academic and production environments.

MLflow

Why MLflow was used

MLflow was chosen for experiment tracking, metric logging, and model registry management. It provides a centralized platform to compare models, track parameters, and manage model versions throughout the lifecycle.

Alternative tools not used

- **Weights & Biases (W&B)** – cloud-based, limited free tier
- **Neptune.ai** – paid enterprise features
- **Comet ML** – subscription-based

Justification

MLflow is fully open source and does not require vendor lock-in. It integrates easily with TensorFlow, scikit-learn, and DVC, making it ideal for transparent, reproducible MLOps workflows.

FastAPI

Why FastAPI was used
FastAPI was chosen to deploy the trained model as a REST API. It supports high-performance asynchronous requests and automatic API documentation.

Alternative tools not used

- **Flask** – slower, manual validation
- **Django REST Framework** – heavy and complex

Justification
FastAPI provides better performance and built-in data validation, making it ideal for real-time ML inference services.

TOOLS AND ARCHITECTURE

Evidently AI

Why Evidently AI was used

Evidently AI was selected for monitoring data drift and prediction drift in production. It provides automated reports and visualization for model monitoring.

Alternative tools not used

- **WhyLabs** – paid service
- **Arize AI** – enterprise-focused
- **Fiddler AI** – closed-source

Justification

Evidently AI is open source, easy to integrate, and well-suited for academic and production MLOps use cases.

TOOLS AND ARCHITECTURE

Docker and GitHub Actions

In the proposed system, **Docker is not used manually**, but is **integrated within the CI/CD pipeline using GitHub Actions**. Docker is invoked automatically by GitHub Actions workflows to build, package, and deploy the machine learning inference service. This integration ensures that every code or model update is consistently containerized and deployed without human intervention.

GitHub Actions triggers Docker build steps whenever changes are pushed to the repository or when a new champion model is promoted. The Docker image encapsulates the FastAPI application, the trained model, and all required dependencies, ensuring environment consistency across development, testing, and production stages. Once built, the container image is deployed to the hosting platform, enabling reliable and repeatable deployments.

Why Docker via GitHub Actions (Not Manual Docker)

Using Docker through GitHub Actions offers several advantages over manual containerization. First, it eliminates configuration drift by ensuring that all deployments are built using the same automated process. Second, it reduces operational overhead by embedding container creation directly into the CI/CD workflow. Third, it improves reproducibility, as each Docker image is versioned and traceable to a specific Git commit and pipeline run.

PHASE 2 – MLOps LIFE CYCLE

Component	MLOps Tools Used
Data Management	DVC for dataset versioning, Apache Spark for scalable preprocessing
Experiment Tracking and Version Control	MLflow for experiment tracking, Git for code versioning
Automation and Pipeline Orchestration	GitHub Actions for CI/CD automation
Model Deployment & Serving	FastAPI for inference API, Docker for containerization, Render for cloud hosting
Monitoring & Governance and Collaboration	Evidently AI for drift detection, SHAP for explainability, MLflow Registry for governance

Data Management

Data management forms the foundation of the entire MLOps lifecycle, as the performance and reliability of machine learning models depend heavily on the quality and consistency of data. In this project, the raw data consists of unstructured user-generated chat messages that include slang, emojis, abbreviations, spelling variations, and informal expressions. Such data is prone to frequent changes over time due to evolving language patterns.

To handle these challenges, **Data Version Control (DVC)** is used to track multiple versions of the dataset. Every update to the dataset, such as the addition of new chat messages or preprocessing changes, is stored as a new version. This ensures that models can always be traced back to the exact dataset used during training, enabling reproducibility and auditability.

Experiment Tracking and Version Control

Once the data is prepared, multiple machine learning and deep learning models are trained to identify toxic content. Since model development involves repeated experimentation with different algorithms, hyperparameters, and preprocessing techniques, systematic tracking is essential.

MLflow is used as the primary experiment tracking tool. For each experiment, MLflow records model parameters, training configurations, evaluation metrics such as accuracy, precision, recall, F1-score, and ROC-AUC, as well as trained model artifacts. This enables detailed comparison across experiments and helps identify the most reliable model.

Final Outcomes

Project Outcomes

- Developed a **production-ready AI Meeting Intelligence Platform** capable of multi-speaker transcription, summarization, and actionable insight generation using a fully reproducible MLOps architecture.
- Implemented a **Level-3 end-to-end MLOps pipeline** with DVC-based data versioning, MLflow experiment tracking and model registry, Dockerized deployment, and automated retraining support.
- **Fine-tuned a domain-specific summarization model** (BART / Long-T5) on meeting-style data, achieving strong ROUGE-1 and ROUGE-L scores while minimizing hallucinations through controlled training and evaluation.
- Built and deployed **in-house trained downstream models** (summarization and sentiment analysis) to demonstrate complete ML lifecycle ownership, from dataset preparation to monitoring and versioned deployment.
- Established **continuous monitoring and governance mechanisms**, including statistical and semantic drift detection, audit-ready model lineage, and PII-aware logging for long-term reliability and compliance.

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THANK YOU